軟體測試

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授課時間：每週五 9：10—12：00 識別碼：921 U9550
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期末考

考試中，請勿參閱資料。

In **Figure 1** is a procedure in C.

```
evenSum(a, b)
   int a, b;
{
      int sum = 0;
      for (int k = a; k <= b; k++)
            if (k % 2 || (k% 5 && k < sum))
                  sum = sum + k;
      if (sum > 2k)
           return (sum);
       else
           return -1;
} // evenSum( )
```

Figure 1: a simple C procedure

1. Draw the control-flow graph of the above program. (5pts/5)

2. List the test requirements for edge-pair coverage for the procedure in Figure 1. (5pts/10)

3. Continuing from question 2, please list a set of test paths for full edge-pair coverage. (5pts/15)

4. Continuing from question 2, list the test requirements for prime-path coverage (PPC). (5pts/20)

5. Continuing from question 2, please write down a test set for the full PPC. (5pts/25)

6. Continuing from question 2, please list all of the du-paths with respect to *k* and *sum*. (Note: Include all du-paths, even those that are subpaths of some other du-path). (5pts/30)

7. Continuing from question 6, please write down a minimal test set that satisfies all-defs coverage with respect to *k* and *sum*. (Direct tours only.) (5pts/35)

8. Continuing from question 6, please write down a minimal test set that satisfies all-uses coverage with respect to *k* and *sum*. (Direct tours only.) (5pts/40)

9. The predicate of the if-statement in **Figure 1** is:

$$(\underline{k \% 2} \mid\mid (\underline{k\% 5} \;\&\&\; \underline{k < sum}))$$

Please identify the clauses (*atoms* in standard logic terms) that determine this predicate. (5pts/45)

10. Continuing from question 9, compute (and simplify) the conditions under which each of the clauses determines the predicate. (5pts/50)

11. Continuing from question 9, write the complete truth table for all clauses. Label your rows starting from 0 at the top. Use the format that row 0 should be all clauses true. You should include columns for the conditions under which each clause determines the predicate, and also a column for the predicate itself. (5pts/55)

12. Continuing from question 9, identify all pairs of rows from your table that satisfy *general active clause coverage* (*GACC*) with respect to each clause. (5pts/60)

13. Continuing from question 9, identify all pairs of rows from your table that satisfy *correlated active clause coverage* (*CACC*) with respect to each clause. (5pts/65)

14. Continuing from question 9, identify all rows from your table that satisfy *implicant coverage* (*IC*) with respect to each implicant. (5pts/70)

15. Assume that we have a web program that collect registration data with the following screen layout.

Figure 2: a GUI for registration and query password.

There is no other document. Please answer the following questions to properly test the page. Please define characteristics of inputs to the web interface and list them. (5pts/75)

16. Continuing from question 15, please partition the characteristics into blocks and write them down. (5pts/80)

17. Continuing from question 16, please define and list values for the blocks. (5pts/85)

18. Assume that we have the procedure in **Figure 1** to test. Please write down a mutant with the Arithmetic Operator Replacement (AOR) and write down a test case (values for *a* and *b*) to kill the mutant. You must explain correctly why the mutant is killed by your test case. (5pts/90)

19. Assume that we have the procedure in **Figure 1** to test. Please write down a mutant with the Scalar Variable Replacement (SVR) and write down a test case (values for a and b) to kill the mutant. You must explain correctly why the mutant is killed by your test case. (5pts/95)

20. Continuing from Question 3, please write down the test cases (as pairs of parameter values) and prioritize them for the best efficiency of edge-pair coverage. (5pts/100)